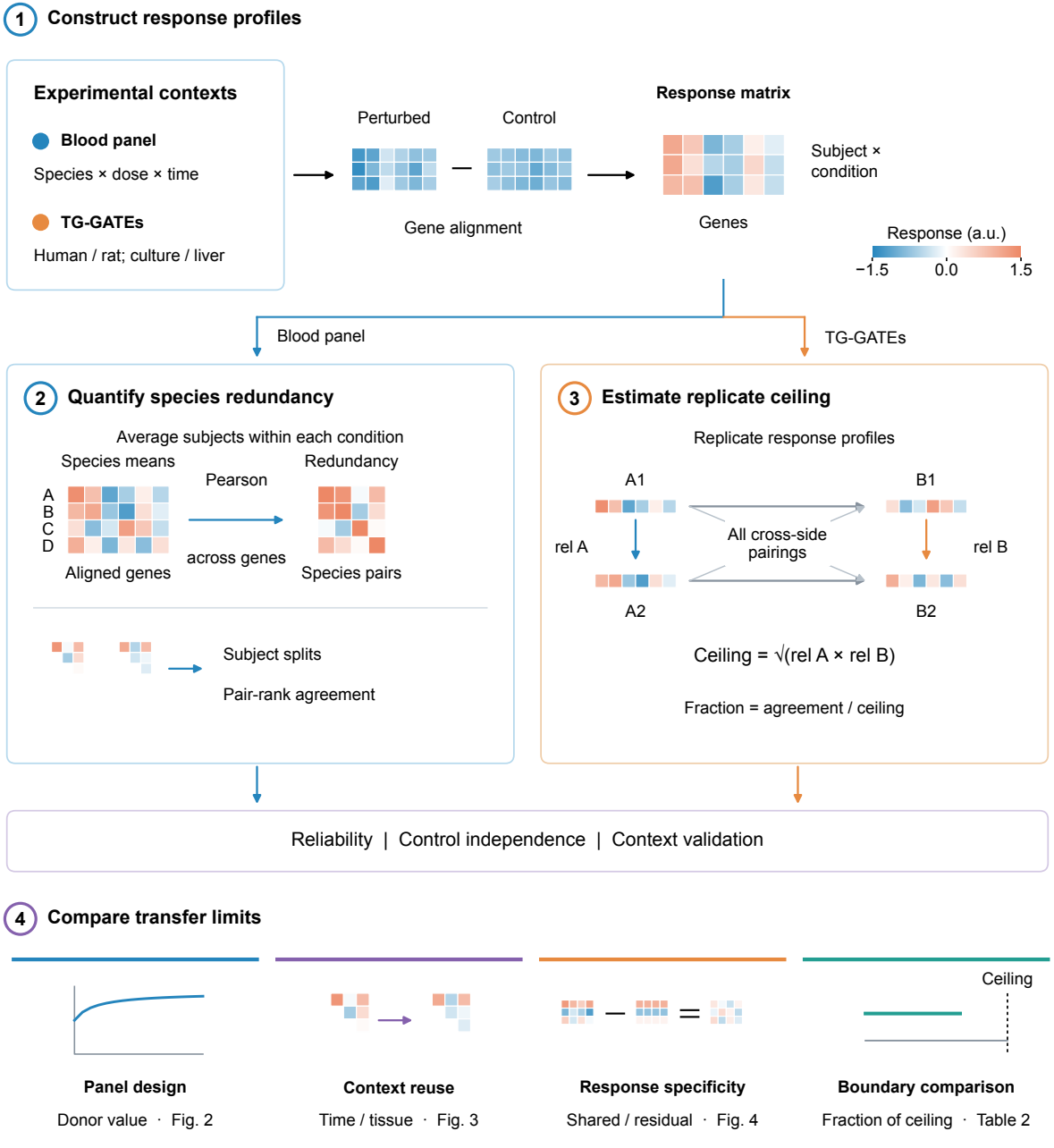

1 Construct response profiles
Experimental contexts
Blood panel
Species × dose × time
TG-GATEs
Human / rat; culture / liver
Perturbed
Control
Gene alignment
Response matrix
Genes
Subject × condition
Response (a.u.)
−1.5
0.0
1.5
Blood panel
TG-GATEs
2 Quantify species redundancy
Average subjects within each condition
Species means
Redundancy
Pearson
across genes
A
B
C
D
Aligned genes
Species pairs
Subject splits
Pair-rank agreement
3 Estimate replicate ceiling
Replicate response profiles
A1
B1
rel A
All cross-side pairings
rel B
A2
B2
Ceiling = √(rel A × rel B)
Fraction = agreement / ceiling
Reliability | Control independence | Context validation
4 Compare transfer limits
Ceiling
Panel design
Donor value · Fig. 2
Context reuse
Time / tissue · Fig. 3
Response specificity
Shared / residual · Fig. 4
Boundary comparison
Fraction of ceiling · Table 2